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A Review of IoMT-based Sleep Apnea Monitoring Systems using Wearable Sensors

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***Abstract*—Obstructive sleep apnea-hypopnea (OSAH) is prevalent sleep-related breathing disorder and is associated with a considerable effect on cardiovascular health and quality of life. Despite the acceptance of polysomnography (PSG) as a clinical gold standard for diagnosis, being expensive, complex to install and unable to provide long-term monitoring restrict its readiness for large-scale implementation. Recent advances in wearable sensors, embedded devices and Internet of Things (IoT) technologies made it possible to propose alternative strategies for the non-invasive and continuous sleep apnea monitoring. This paper, however is focusing on the survey for several wearable and IoT based sleep apnea monitoring systems available in literatures published recently. Furthermore, physiological sensor modalities like sensing ECG, blood O_2 saturation (SpO_2), respiratory effort and MEMS-based motion sensors are discussed in detail. Furthermore, system architectures and data processing approach as well as cloud-supported monitoring concept are presented and discussed. Critical challenges in Imaging Wireless Capsule Endoscopy: motion artifact, power consumption, data integrity and clinical problems are addressed.**

Keywords— Obstructive Sleep Apnea (OSA), Wearable Sensors, Internet of Medical Things (IoMT), ECG and SpO_2 Monitoring, MEMS Motion Sensors, Secure UART Communication, Remote Health Monitoring.

I. INTRODUCTION

Sleep apnea is a common sleep disorder that causes intermittent hypoxia and fragmented sleep by repeatedly stopping or reducing airflow during sleep. The most common type, obstructive sleep apnea (OSA), is closely associated with heart arrhythmias, stroke, hypertension, and a lower quality of life. Due to the reliance on polysomnography, which necessitates specialized sleep laboratories and qualified clinicians, a significant portion of affected individuals remain undiagnosed despite its clinical significance[5]. Wearable sensor technology and Internet of Things-based healthcare systems have received a lot of attention lately for monitoring sleep disorders. These systems are designed to offer home-based, portable, and reasonably priced monitoring options. This review examines current wearable and Internet of Things-enabled sleep apnea monitoring methods, evaluating their performance metrics, sensing modalities, and system design strategies[6].

II. SLEEP APNEA OVERVIEW

The gold standard for diagnosing sleep apnea is polysomnography, which records several physiological signals at once, including respiratory effort, oxygen saturation, airflow, EEG, and ECG. PSG has disadvantages despite its high accuracy, such as high cost, patient discomfort, restricted accessibility, and incapacity to facilitate long-term monitoring. Research into alternate wearable and Internet of Things-based solutions has been spurred by these difficulties [7].

III.SENSOR SPECIFICATIONS

A. ECG

In order to identify autonomic nervous system reactions brought on by apnea episodes, ECG-based systems examine heart rate and heart rate variability (HRV). ECG is a useful non-invasive indicator because several studies show that apnea episodes are correlated with distinctive HRV patterns[8].

B.SpO_2

One of the most popular metrics for identifying sleep apnea is blood oxygen saturation (SpO_2). Apnea and hypopnea episodes can be directly detected by oxygen desaturation events. Home-based monitoring systems frequently incorporate wearable pulse oximeters[9].

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C.MEMS

MEMS gyroscopes and accelerometers are used to track sleeping posture and body movement. These sensors are especially helpful in identifying positional sleep apnea and lowering the number of false positives brought on by motion artifacts[10].

IV.SYSTEM BASED IOT ARCHITECTURE

A wearable sensing unit, an embedded processing module, and a cloud or mobile-based monitoring interface are the usual components of IoT-enabled sleep apnea monitoring systems. Because of their low power consumption, built-in Wi-Fi, and real-time data handling capabilities, microcontrollers like the ESP32 are widely used. IoT platforms give clinicians and patients access to data storage, alert systems, and remote monitoring.

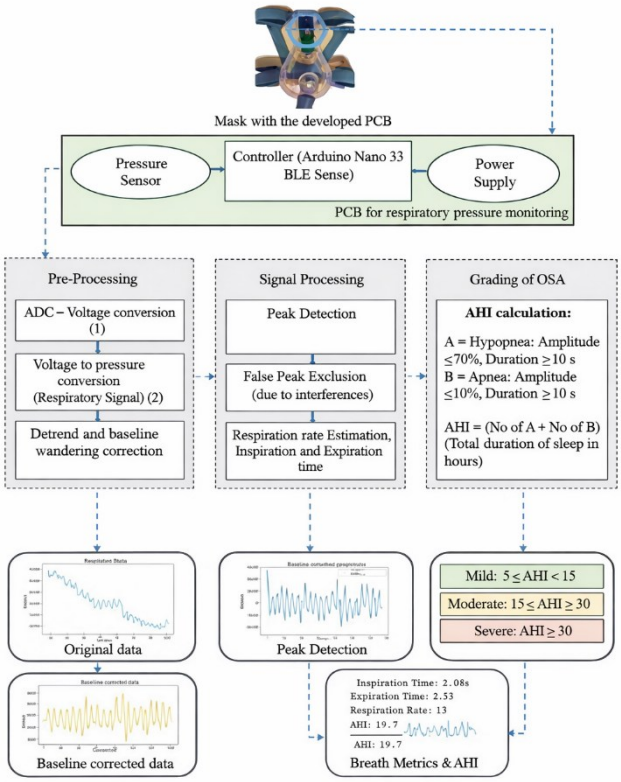


Fig 1: Previous paper methodology

V.COMPARATIVE ANALYSIS

IoMT-based and wearable solutions have received more attention recently as alternatives to traditional polysomnography for sleep apnea monitoring. In order to demonstrate the feasibility of ongoing monitoring outside of clinical settings, Dhruba et al. [1] created an Internet of Things (IoT)-based sleep apnea monitoring system for healthcare applications that allows real-time physiological data acquisition and remote observation, albeit with a greater reliance on cloud connectivity. Sch□ et al. [2] improved the accuracy of apnea detection without direct body contact in a clinical setting by combining polysomnography with depth sensors; however, the system is still expensive, non-portable, and unsuitable for long-term home-based monitoring.

From Fig [1] Recent wearable-focused research, however, places more emphasis on portability and non-invasive use. A novel wearable technology for the non-invasive treatment of positional sleep apnea was proposed by Leogrande et al. [3].

suggested a novel wearable technology that focuses more on therapeutic intervention than full apnea detection for the non-invasive treatment of positional sleep apnea. In the meantime, Delpha Jacob et al. [4] demonstrated the efficacy of multi-sensor wearable platforms for ongoing screening by presenting a wearable sensor design for real-time obstructive sleep apnea monitoring. Despite these developments, there is still a need for more reliable and energy-efficient designs because current IoMT-based wearable systems still have issues with motion artifacts, power efficiency, and limited large-scale clinical validation shown in fig.2.

Ref.	Approach	Key Sensors	Main Focus
[1]	IoMT-based system	ECG, SpO_2	Remote real-time monitoring
[2]	Clinical PSG + depth sensing	PSG, depth sensor	High-accuracy clinical detection
[3]	Wearable therapeutic device	Motion / position	Positional apnea management
[4]	Wearable real-time system	Multi-sensor wearable	Continuous apnea monitoring

Fig 2: Comparison of existing papers

VI.PROPOSED SYSTEM

From Fig [3] The suggested IoMT-based sleep apnea monitoring system is intended to be a wearable, non-invasive, and real-time monitoring tool. In order to record heart rate, blood oxygen saturation, and body movement while you sleep, the system incorporates physiological sensors such as ECG and SpO_2 modules in addition to a MEMS motion sensor. An ESP32 microcontroller, which acts as the central processing unit and handles data collection, initial signal processing, and secure communication coordination, is interfaced with these sensors.

UART, which is improved with lightweight encryption and checksum verification to guarantee the confidentiality, integrity, and dependability of sensitive patient data, is used to transfer sensor data to the ESP32. The synchronised multi-sensor data is further processed by the ESP32 to identify possible apnea events in real time. It is appropriate for continuous overnight monitoring in home environments

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because secure UART communication reduces latency and energy consumption while preserving system simplicity.

Healthcare professionals or carers can access real-time physiological data and alerts for remote monitoring and visualization by sending the processed data via the IoMT framework to a cloud platform or mobile application. This integrated architecture offers a reliable, low-power, and privacy-preserving sleep apnea monitoring system by combining small wearable hardware, secure serial communication, and IoMT-enabled data delivery.

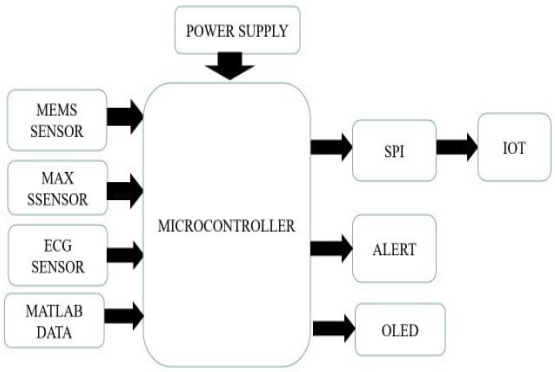


Fig 3: Block diagram of proposed system

VII. SECURE UART COMMUNICATION

UART (Universal Asynchronous Receiver–Transmitter) is used as the main serial interface between wearable physiological sensors and the embedded processing unit in the suggested IoMT-based sleep apnea monitoring system. Because of its ease of use, low hardware requirements, and dependable point-to-point communication, UART is used to transmit continuous biomedical signals generated by ECG, SpO_2, and motion sensors. Compact wearable architectures benefit greatly from UART's asynchronous nature, which enables reliable sensor-to-controller data transfer Lightweight encryption and checksum verification are used in secure UART communication to guarantee the confidentiality and integrity of sensitive patient data. Sensor data packets are encrypted by the embedded controller prior to transmission, and the receiving module verifies and decrypts them in real time. For healthcare IoMT applications where patient privacy and data security are crucial, this guarantees that biomedical signals are shielded from interception, tampering, or data corruption.

Additionally, incorporating secure UART facilitates real-time processing for sleep apnea detection and allows for effective synchronisation of multi-sensor data streams. The system achieves both energy efficiency and secure operation by integrating encryption and validation protocols with low-overhead asynchronous communication. For dependable, private data transfer in wearable sleep apnea monitoring devices, this method offers a workable solution.

UART Transmission Equation

UART sends serial bits:

$$\text{Frame} = [\text{Start bit}] + [\text{Data bits}] + [\text{Parity bit}] + [\text{Stop bit}]$$

For secure UART:

- Data bits = Encrypted sensor byte
- Parity / checksum = integrity check

So the effective secure UART frame:

$$\text{Secure Frame} = \text{Start} + E_K(D) + \text{Checksum}(C) + \text{Stop}$$

VIII. DATA VISUALIZATION AND USER INTERFACE

The Blynk IoT platform and MATLAB for real-time visualization and analysis are used in the dual user interfaces of the suggested sleep apnea monitoring system. The main user interface for remote monitoring is the Blynk IoT application, which displays physiological parameters like heart rate, SpO_2 levels, and motion status in real time. These parameters are sent from the wearable device via the IoMT framework. Users and carers can monitor patient conditions in real time and receive alerts when abnormal thresholds suggestive of apnea events are detected thanks to the mobile-based interface.

MATLAB is used as an analytical user interface for offline and semi-real-time signal visualization in addition to mobile monitoring. Plotting user physiological data from the embedded system in MATLAB allows for the observation of motion patterns, oxygen saturation trends, and ECG waveforms over time. Detailed signal-level analysis, apnoea detection logic validation, and performance evaluation of the suggested system under various sleep conditions are all made easier by this graphical representation. Blynk IoT and MATLAB together give the system accessibility and analytical depth. MATLAB offers sophisticated visualisation and research-focused analysis, while Blynk guarantees user-friendly real-time monitoring and alert generation appropriate for IoMT-based healthcare applications. In wearable sleep apnoea monitoring applications, this dual-interface strategy improves system usability, dependability, and clinical relevance shown in fig.4.

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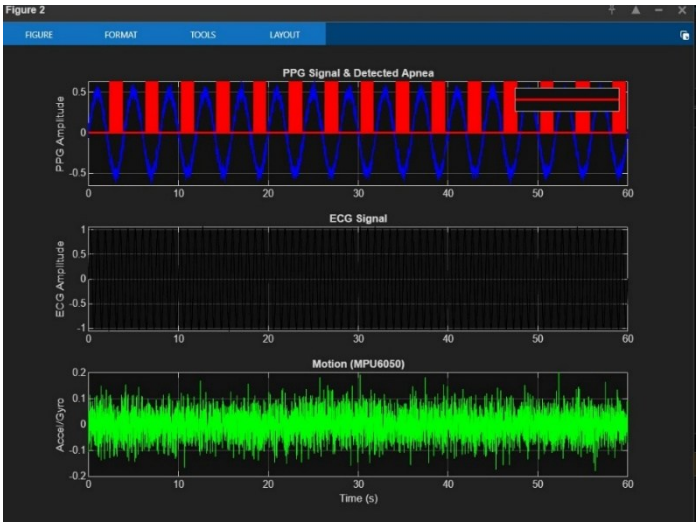


Fig 4: Simulated output using MATLAB

IX. CONCLUSION

In addition to a safe and effective system architecture for real-time monitoring, this paper provided a thorough analysis of IoMT-based wearable sensor-based sleep apnea monitoring systems. The benefits and drawbacks of current methods utilizing wearable technology, polysomnography, and IoMT frameworks in terms of portability, power consumption, robustness, and clinical applicability were examined. According to the review, wearable and IoMT-based solutions offer viable substitutes for traditional diagnostic methods, especially for long-term and home-based monitoring. To guarantee dependable and private data transfer between wearable sensors and the embedded processing unit, a secure UART-based communication framework was incorporated into the suggested system. In order to infer possible apnea events, the system also uses decision logic, real-time preprocessing, and multi-sensor data acquisition. System usability and analytical capability are improved by integrating MATLAB for detailed signal visualization and Blynk IoT for real-time remote monitoring. All things considered, the suggested IoMT-based wearable architecture provides a scalable, economical, and energy-efficient sleep apnea monitoring solution. This work advances the development of useful and user-friendly wearable healthcare monitoring systems appropriate for continuous sleep apnea assessment by tackling important issues like secure data communication, real-time visualization, and system integration.

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